Investment Thesis – Viking Therapeutics

Company Overview

Viking Therapeutics (Viking) is a clinical-stage biopharmaceutical company that is focused on developing novel drugs for metabolic and endocrine disorders. Currently, Viking has three compounds in clinical trials for the treatment of obesity (VK2735), non-alcoholic steatohepatitis (NASH) – an advanced stage of non-alcoholic fatty liver disease (VK2809), and X-linked adrenoleukodystrophy (X-ALD) – a rare genetic neurodegenerative disorder that affects the spinal cord, brain, and adrenal glands (VK0214). Viking trades on the NASDAQ stock exchange under the ticker VKTX and is currently focusing on FDA approval for its lead drug candidates. VK0214 is omitted from this analysis since it targets the small X-ALD treatment market and contributes minimally to Viking's overall value.

VK2735

VK2735 is Viking's highest potential drug asset it its development pipeline, targeting obesity which is a widespread and chronic metabolic disorder with growing prevalence. It is a novel glucagon-like peptide (GLP-1)/glucose dependent insulinotropic polypeptide (GIP) dual receptor agonist in both subcutaneous and oral formulations, aiming to improve weight loss while offering better gastrointestinal (GI) tolerability and multiple methods of administration.

The most recent clinical trial for the subcutaneous formulation was Phase 2 VENTURE, where VK2735 met its primary endpoint with statically significant weight loss of up to -14.7% at 13 weeks for treated patients compared to just -1.7% for the placebo group. The drug was well tolerated with 92% of reported adverse events being categorized as mild or moderate. For the oral formulation, it demonstrated a promising safety profile in its Phase 1 trial, enabling the initiation of Phase 2 trials in January 2025.

The GLP-1 obesity treatment market in the US is rapidly growing and is projected to reach $74 billion USD by 2036. Despite competition from incumbents like Novo Nordisk and Eli Lilly, VK2735 is well positioned to capture market share since it would be the first GLP-1/GIP dual receptor agonist with an oral option. Based on conservative assumptions, estimated total peak sales of $4.4 billion USD for both formulations imply a valuation range of $15.86 to $25.17 USD per share for the subcutaneous formulation and $5.75 to $9.12 USD per share for the oral formulation. With its competitive advantage, massive market and promising clinical results, VK2735 is the major value driver of Viking. *A full breakdown of this analysis is provided in the Analysis of VK2735 section below.*

VK2809

VK2809 is a high-potential drug asset in Viking's development pipeline, targeting NASH which is a high-prevalence and high-burden liver disease with limited treatment options. The drug is designed as a liver-selective thyroid hormone receptor beta (TRβ), meaning it reduces hepatic fat (liver fat) and improves lipid metabolism while mitigating the risk of cardiovascular side effects associated with the activation of the other major TR.

In its most recent clinical trial, Phase 2b VOYAGE, VK2809 demonstrated its efficiency by meeting both its primary and secondary endpoints with statistical significance when compared to the placebo group. Notably, at the 12-week mark, patients treated with VK2809 demonstrated median reductions in hepatic fat ranging from 38% to 55% from baseline, whereas it was just 5.4% for the placebo group. VK2809 also demonstrated strong safety and tolerability.

Currently, the only FDA-approved drug for NASH treatment is Rezdiffra of Madrigal Pharmaceuticals, indicating that VK2809 faces low near-term competition. Based on conservative assumptions, estimated peak

sales of $1.235 billion USD imply a valuation range of $8.90 to $14.13 per share. With its clinical profile and market opportunity, VK2809 is a key value driver of Viking. *A full breakdown of this analysis is provided in the Analysis of VK2809 section below.*

Valuation

To ascertain a valuation range for Viking, a probability-adjusted net present value (pNPV) analysis (see pNPV) was conducted since Viking is a clinical-stage biopharmaceutical company with no current revenues. Applying this framework to Viking's lead drug candidates VK2735 and VK2809 yields an implied price per share of $35.09 USD, as illustrated in Table 1.

Table 1. Probability-adjusted net present value per share of Viking's pipeline

Drug	Method of Administration	Indication	Status	Lanch Year	Success	Peak Sales (USDm)	Profitability	Probability Weighted Peak Profit (USDm)	pNPV (USD)
VK2735	Subcutaneous	Obesity	Phase 3	2031	60%	2,200	40%	528	18.24
VK2735	Oral	Obesity	Phase 2b	2032	25%	2,200	40%	220	6.61
VK2809	Oral	NASH	Phase 3	2031	60%	1,235	40%	296	10.24
								Total	**35.09**

To estimate a valuation range for Viking, a sensitivity analysis was performed by varying both the discount rate and price to earnings (P/E) ratio built into the model. As discussed in the commercial model below, a discount rate of 15% and a P/E ratio of 18 are assumed. Adjusting these inputs at the lower and upper bounds of the combined valuations for VK2735 and VK2809 provides a comprehensive view of the asymmetric upside. Summing the valuation ranges for each drug yields a lower bound of $30.51 USD per share and an upper bound of $48.42 USD per share. These values serve as the basis for the sensitivity analysis shown in Tables 2 and 3.

Table 2. Lower bound sensitivity analysis for implied value per share of Viking

		P/E ratio				
		16	17	18	19	20
	11%	41.76	44.37	46.98	49.59	52.20
	13%	33.59	35.69	37.79	39.88	41.98
Discount rate	**15%**	27.12	28.82	30.51	32.21	33.90
	17%	21.98	**23.35**	24.73	26.10	27.48
	19%	17.88	19.00	20.11	21.23	22.35

Table 3. Upper bound sensitivity analysis for implied value per share of Viking

		P/E ratio				
		16	17	18	19	20
	11%	61.74	65.60	69.46	73.31	77.17
	13%	51.46	54.68	57.90	**61.11**	64.33
Discount rate	**15%**	43.04	45.73	48.42	51.11	53.80
	17%	36.11	38.36	40.62	42.88	45.13
	19%	30.38	32.28	34.18	36.08	37.98

Given that the P/E ratio for mature revenue-producing pharmaceutical companies has historically remained around 16-18 [1], which Viking is assumed to mature into in the pNPV analysis, a range of 17-19 centered around the base case of 18 is used. The discount rate is varied by $\pm$ 2% from the base case of 15%, with a cap of 17% since peak sales are already risk-adjusted. Taking the lowest value from the lower bound in Table 2 and the highest value from the upper bound in Table 3 yields a final valuation range for Viking of $23.35 to $61.11 USD per share, with a base case target price of $35.09 USD per share.

Recommendation

Viking presents a wonderful investment opportunity, driven by its two promising late-stage drug assets VK2735 and VK2809 targeting obesity and NASH, both of which are high-prevalence and chronic conditions. Each drug has advanced passed Phase 2 trials with statistically significant clinical results and impressive safety profiles, moving closer to commercialization in large and growing markets. VK2735 is well positioned to compete in the rapidly expanding obesity treatment market, offering both subcutaneous and oral formulations, and differentiating itself from incumbents as potentially becoming the first FDA-approved oral GLP-1/GIP dual receptor agonist. VK2809 benefits from a strong drug profile and minimal near-term competition in the NASH treatment market.

Applying a pNPV analysis to Viking's pipeline yields a base case target price of $35.09 USD per share, offering an upside of 32% from its price of $26.67 as of the day of this writing (July 1st, 2025). The sensitivity analysis suggested a valuation range of $23.35 to $61.11 USD per share, reflecting a meaningful upside of up to 129% with limited downside of just 12%. Given the robust Phase 2 clinical results, unique drug differentiation, large market opportunities and the asymmetric upside, Viking is recommended as a Buy rating.

Limitations and Next Steps

The primary limitation of this analysis is the static nature of the pNPV method and its failure to model management's flexibility in deciding to continue or stop the development of a drug asset at each stage. Drug development is a multi-year process, and at any stage, the drug may not be suitable to advance to the next phase of development, thus reducing the company's value. The pNPV framework assumes management makes a single decision to continue with the drug's development at the time of valuation and does not build in the flexibility to end development at a later stage.

A natural next step is to build in a layer of sophistication by incorporating real options analysis to model management's flexibility throughout the development timeline more accurately. Real options capture management's flexibility of allocating substantial capital to development stages only when clinical results are promising and forgoing investment when results are poor. Another improvement comes from building a more robust market model that accounts for population growth by indication, stratifies patients by disease severity, incorporates diagnosis rates, and estimates new patients and switching patients. This would provide a more comprehensive picture of the potential patient market share captured by the drug and its competitors. Finally, factoring in Viking's cash on hand and their burn rate would determine how long current spending on research and development can be sustained before additional capital must be raised.

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Methodology

Probability-Adjusted Net Present Value

Since Viking is a clinical stage biopharmaceutical company with no revenues, its value is derived from its drug pipeline. To value drugs in development, the pNPV method is used to determine the value of VK2735 and VK2809, Vikings' lead drug candidates in their metabolic disease programs. This method relies on the following inputs: peak sales, launch year, number of years to peak sales after launch, overall probability of success, royalty structure, profitability, discount rate, P/E ratio, and the number of shares outstanding.

For each drug asset, peak sales are estimated and multiplied by gross margin (profitability) and a royalty rate (if applicable). This yields the peak profit for the drug asset, which is then multiplied by the drug's overall probability of success based on its current stage of development, yielding the probability-adjusted peak profit. Then, the NPV of these profits is calculated by discounting it over the number of years to peak sales using a defined discount rate. Finally, the NPV of the profits is multiplied by the P/E ratio and divided by the number of shares outstanding to yield the implied price per share. The method is summarized in the following formulas:

$$Implied\ valuation = \frac{pNPV \times P/E}{Number\ of\ shares\ outstanding}\ ;\ pNPV = \frac{Peak\ sales \times Profitability \times Royalty\ rate \times Probability\ of\ success}{(1 + Discout\ Rate)^{Years\ to\ launch\ +\ Years\ to\ peak\ sales\ after\ launch}}$$

Commercial Model

The purpose of developing a commercial model is to estimate the NPV of VK2809 by discounting its peak sales to reflect today's value. This analysis requires the following variables: launch year, number of years to peak sales after launch, overall probability of success, royalty structure, profitability, discount rate, P/E ratio, and the number of shares outstanding.

- **Launch Year.** Estimating launch year requires the consideration of typical development timelines. For orally administered drugs, Phase 2 and Phase 3 trials last an average of 28.31 and 44.88 months, respectively. For subcutaneously administered drugs, Phase 3 trials last an average of 49.86 months. In general, Phase 3 trials are followed by an additional 18.2 months for regulatory review and approval [1]. Since VK2809 has completed Phase 2, its launch year is estimated to be 2031 if successful. For the oral formulation of VK2735, its launch year is expected to be 2032 since Phase 2 trials were initiated in January 2025; and for the subcutaneous formulation, its launch year is expected to be 2031 since it has completed Phase 2.
- **Number of years to peak sales after launch.** Based on data from expected life cycle sales of new drugs, peak sales are typically reached around 5 years post-launch [1].
- **Overall probability of success.** The overall probability of success that a drug entering Phase 2 or Phase 3 will make it to market is approximately 25% and 60%, respectively [1].
- **Royalty structure.** Viking holds exclusive worldwide rights to VK2809 via their licensing agreement with Ligand Pharmaceuticals. Viking will be paying royalties to Ligand rather than receiving them, therefore, royalties are not considered for this valuation. Moreover, VK2735 is owned and internally developed by Viking, so they retain full global commercial rights to the drug. Therefore, royalties are not considered for this asset either.
- **Profitability.** Margins associated to individual drugs (not the entire pharmaceutical company) typically vary between 20% and 60% [1]. A midpoint estimate of 40% is taken.
- **Discount rate.** A 15% discount rate is assumed, which is a standard discount rate used for pharmaceutical pNPV analysis [1].
- **P/E ratio.** The P/E ratio used is the typical one applied to pharmaceutical companies. The assumption is that Viking will ultimately mature and resemble a pharmaceutical company that makes profit. As of January 2025, data from NYU Stern indicates a forward looking P/E ratio of 18.72 for the pharmaceutical industry [2]. To be conservative, a P/E ratio of 18 is used.
- **Number of shares outstanding.** From 2024 to 2025, the weighted average number of shares outstanding for Viking is 112 million.

Analysis of VK2735

Disease Overview

The development of VK2735 targets obesity, which is a chronic and highly prevalent metabolic disorder associated with increased risk of type 2 diabetes, cardiovascular disease, musculoskeletal disorders and cancer [3]. Obesity is characterized by excessive fat accumulation due to greater energy intake compared to expenditure. From August 2021 to 2023, the prevalence of obesity among adults in the US was 40.3%, making it a major contributor to mortality, morbidity and healthcare costs [4]. Over the last decade, GLP-1 receptor agonists and GLP-1/GIP dual agonist therapies have emerged to treat obesity. However, with the incidence of obesity growing, there remains unmet need for therapies that improve weight loss while offering better safety and tolerability.

Drug Profile

VK2735 is a novel GLP-1/GIP dual receptor agonist in development for both subcutaneous and oral formulations. GLP-1 and GIP are incretin hormones, which are gut hormones released after eating that enhance glucose-dependent insulin secretion and improve glycemic control [5]. When GLP-1 is secreted, it binds to GLP-1 receptors in the pancreas, central nervous system and stomach. The downstream effects are stimulated insulin secretion and suppressed glucagon in the pancreas, slowed gastric emptying in the stomach and reduced appetite via pathways in the central nervous system [6].

Glucagon is a hormone that breaks down glycogen into glucose in the liver, resulting in higher blood sugar, and insulin is a hormone that lowers blood sugar by facilitating cellular uptake for energy storage [6]. In obesity, people have chronically elevated glucagon levels, raising blood sugar and triggering increased insulin release. This hormonal imbalance results in continuous fat storage and inhibits lipolysis (fat breakdown). Targeted incretin therapies like GLP-1 agonists stimulate insulin secretion only when blood glucose levels are elevated. They also simultaneously suppress glucagon, reducing glucose output from the liver. This dual action reduces blood sugar spikes, thereby improving insulin sensitivity and shifting the hormonal balance towards reduced fat storage and increased lipolysis [6].

GIP secretion also enhances glucose-stimulated insulin secretion, though it does not suppress glucagon. The important thing to note for VK2735 is that when GIP receptor agonists are combined with GLP-1 receptor agonists as a dual GLP-1/GIP agonist, it shows improved therapeutic benefits while offering less GI side effects [5]. By engaging both receptors, VK2735 aims to achieve greater weight loss benefits while minimizing side effects, emphasizing its position as a promising candidate in the quickly expanding obesity therapeutic market.

Latest Clinical Data

Subcutaneous Formulation – Phase 2

The most recent clinical trial conducted for the subcutaneous formulation of VK2735 was Phase 2 VENTURE, a randomized, double-blind, placebo-controlled study on 176 adults who are obese or overweight. The objectives of the study were to assess the drug's efficacy at various dosing levels, and its safety and tolerability. The primary endpoint evaluated the percentage change in body weight from baseline at week 13, and secondary and exploratory endpoints assessed safety, tolerability and other efficacy measures. Patients treated with VK2735 received weekly injections of either 2.5 mg, 5 mg, 10 mg or 15 mg [7].

For the primary endpoint, patients treated with VK2735 demonstrated statistically significantly mean percent change in body weight from baseline at week 13 of -9.1% to -14.7%. By contrast, patients in the placebo group experienced a mean percent change of just -1.7%, meaning that the placebo-adjusted mean percent change was -7.4% to -13.1%. Additionally, 39% to 88% of patients treated with VK2735 experienced a mean body weight loss of at least -10% compared to just 4% of patients in the placebo group. Finally, the weight loss

was progressive for treated patients, with no plateauing after 13 weeks, suggesting additional weight loss may be achieved following the 13 weeks [7].

Regarding the safety and tolerability, 92% of reported drug treatment emergent adverse events (TEAEs) were described as either mild or moderate, with 95% of these TEAEs being GI [7]. Additional information presented during Obesity Week in 2024 showed that patients treated with VK2735 maintained 83% of weight loss seven weeks after receiving the last dose [8].

Oral Formulation – Phase 1

The most recent clinical trial conducted for the oral formulation of VK2735 was Phase 1 multiple ascending dose, a randomized, double-blind, placebo-controlled study on 47 healthy adults. The objective of the study was to assess the safety and tolerability of the drug. The oral formulation demonstrated promising results following the 28-day trial of once-daily dosing, with all TEAEs being mild or moderate. From these results, Viking initiated a 13-week Phase 2 trial for the oral formulation in January 2025 [9].

Market Analysis

The following analysis estimates peak sales for VK2725 based on a top-down model. With the target market being the US, key inputs for the analysis include competitors in the US, the obesity treatment US market size, and the US market share of VK2735.

- **Competitors.** The primary competition in the GLP-1 receptor agonist market (which includes GLP-1/GIP dual receptor agonists) is driven by Novo Nordisk and Eli Lilly. Novo Nordisk is the leader for GLP-1 receptor agonist therapies, with brands such as Ozempic, Wegovy and Saxenda dominating the market for obesity. In contrast, Eli Lilly is the only company with FDA-approved dual GLP-1/GIP receptor agonist therapies, with Zepbound being the brand for which the obesity drug is sold under. These drugs are all administrated subcutaneously. It is also important to note that Eli Lilly has a GLP-1/GIP/glucagon receptor triagonist (Retatrutide) in Phase 3 for obesity treatment.
- **Obesity treatment US market size.** According to TD Cowen, global sales of the GLP-1 receptor agonist market are projected to reach $139 billion USD by 2030, representing a 21% compounded annual growth rate (CAGR) from 2023-2030 [10]. This captures both type 2 diabetes and obesity therapies, yet they estimate that obesity therapies could capture 42% of the entire market. Based on the commercial model discussed above, VK2735 is expected to reach peak sales in 2036 (subcutaneous) and 2037 (oral), allowing 6-7 years of market growth until peak sales. Assuming a much lower CAGR of 10% from 2030 to 2036 yields an estimated global obesity treatment market of $103 billion USD in 2036. The US accounted for 72.3% of the global GLP-1 receptor agonist market in 2023 and is expected to grow faster than the global average from 2023 to 2030 [11], so a conservative assumption is to maintain this proportion of total sales in 2036. Therefore, the obesity treatment US market is estimated to be $74 billion USD in 2036.
- **Market share.** By 2036, the obesity treatment market will remain highly competitive with leaders like Novo Nordisk and Eli Lilly expected to be in stronger positions. Next generation drugs like Eli Lilly's Phase 3 triagonist Retatrutide, will further increase competition. However, VK2735 offers novelty as a dual agonist with improved GI tolerability and as the first oral dual receptor agonist for obesity treatment. Balancing the competitive market with the novel edge of VK2735, a 6% US market share is assumed in this analysis.

In 2036, the US obesity market size for GLP-1 receptor agonists is estimated to be $74 billion USD with VK2735 capturing 6% of the market. This implies VK2735 peak sales total $4.4 billion USD, which is assumed to be evenly split between its subcutaneous and oral formulations.

Valuation Model

Given the assumptions stated in the commercial model and market analysis, the probability-adjusted peak profit for VK2735 is projected to be $528 million USD in 2036 for the subcutaneous formulation and $220 million USD in 2037 for the oral formulation (Table 1). Applying the pNPV framework outlined above results in an

implied valuation of $18.24 USD per share for the subcutaneous formulation and $6.61 USD per share for the oral formulation.

Table 4. Sensitivity analysis for pNPV per share of VK2735 - subcutaneous

		Years to peak sales				
		9	10	11	12	13
	1320	14.47	12.59	10.94	9.52	8.27
	1760	19.30	16.78	14.59	12.69	11.03
Peak sales (USDm)	**2200**	24.12	20.98	**18.24**	15.86	13.79
	2640	28.95	25.17	21.89	19.03	16.55
	3696	40.52	35.24	30.64	26.65	23.17

Table 5. Sensitivity analysis for pNPV per share of VK2735 - oral

		Years to peak sales				
		10	11	12	13	14
	1320	5.24	4.56	3.97	3.45	3.00
	1760	6.99	6.08	5.29	4.60	4.00
Peak sales (USDm)	**2200**	8.74	7.60	**6.61**	5.75	5.00
	2640	10.49	9.12	7.93	6.90	6.00
	3696	14.68	12.77	11.10	9.65	8.39

To obtain a valuation range, a sensitivity analysis was performed by varying both the peak sales and years to peak sales, as shown in Tables 4 and 5. Given the conservative peak sales estimate of $2.2 billion USD for each formulation, the range considered for valuation spans from this figure to 20% higher or $2.64 billion USD. Since the average development timelines were used to estimate time to peak sales, the sensitivity range adjusts the value by $\pm$ 1 year, resulting in a window of 10 to 12 years for subcutaneous formulation and 11 to 13 years for the oral formulation. Under these scenarios, the current valuation of VK2735 ranges from $15.86 to $25.17 USD per share for the subcutaneous formulation and $5.75 to $9.12 USD per share for the oral formulation.

Analysis of VK2809

Disease Overview

The development of VK2809 targets NASH, which is a progressive and severe form of non-alcoholic fatty liver disease. NASH is characterized by hepatic fat accumulation, inflammation and liver cell damage, which over time, can lead to fibrosis, cirrhosis, liver cancer, or liver failure [12]. NASH is both a high-prevalence and high-burden chronic disease, with an estimated overall prevalence of 1.5% to 6.45% of the population [13]. In March 2024, the FDA approved Rezdiffra (resmetirom) of Madrigal Pharmaceuticals, making it the first and only therapy for NASH, emphasizing the significant unmet clinical need in this therapeutic area [14].

Drug Profile

VK2809 is a novel and orally available small molecule TRβ agonist that is selective for liver tissue. This liver selectivity is important since TRs are distributed across various tissues and elicit different physiological responses. There are two major isoforms of TRs: TRβ, which is mostly expressed in the liver, and TRα, which is mainly expressed in the heart and skeletal muscle. While both TRs are activated by the natural thyroid hormone T3, their downstream effects differ [15].

When TRβ is activated, it upregulates genes involved in lipid metabolism. This includes genes that regulate cholesterol clearance and fatty acid oxidation. For NASH, this is crucial because it reduces low-density lipoprotein (LDL) cholesterol and enhances oxidation of fatty acids, meaning that hepatic fat accumulation is reduced which is a key driver of disease progression [15]. In contrast, TRα activation increases cardiac output and heart rate, which can lead to cardiac hypertrophy – an enlargement of the heart that impairs its blood pumping efficiency – and arrhythmias, potentially causing palpitations, stroke or cardiac arrest [15]. Therefore, selective activation of TRβ is essential for effective and safe treatment of NASH, while reducing cardiovascular risks associated to TRα activation.

Latest Clinical Data – Phase 2b

The most recent clinical trial conducted for VK2809 was Phase 2b VOYAGE, a randomized, double-blind, placebo-controlled study on 229 adults with biopsy-confirmed NASH and fibrosis. It was carried out to evaluate the efficacy, safety and tolerability of VK2809. The primary endpoint evaluated change in hepatic fat content after 12 weeks of treatment (measured by magnetic resonance imaging-proton density fat fraction – MRI-PDFF), and the secondary endpoints assessed histologic changes after 52 weeks of treatment. Patients treated with VK2809 received daily doses of either 1 mg or 2.5 mg, or alternate-day doses of 5 mg or 10 mg [16].

Regarding the primary endpoint, patients treated with VK2809 demonstrated statistically significant median reductions in hepatic fat ranging from 38% to 55% from baseline at week 12. By contrast, patients in the placebo group experienced a median reduction of only 5.4% over the same period. Moreover, 53% to 85% of patients treated with VK2809 achieved at least a 30% reduction in hepatic fat from baseline at week 12, which is a threshold associated with a greater likelihood of histologic response in NASH. This proportion of patients was significantly higher than the 13.6% observed in the placebo group, further supporting the therapeutic potential of VK2809 [16].

For the secondary histologic endpoints, VK2809 showcased statically significant improvements at week 52 compared to the placebo group, supporting the efficacy suggested by the primary outcome. In terms of safety and tolerability, 94% of treatment-related adverse events were reported as mild or moderate. Additionally, VK2809 demonstrated strong GI tolerability, with rates of GI-related adverse events being similar among patients treated with VK2809 and those in the placebo group [16].

Market Analysis

The following analysis estimates peak sales for VK2809 based on a bottom-up model. With the target market being the US, key inputs for the analysis include competitors in the US, treatable-eligible NASH population in the US, patient market share of VK2809, and annual revenue per patient for VK2809.

- **Competitors.** Currently, the only FDA-approved therapy for NASH is Rezdiffra, which is a selective TRβ agonist like VK2809. This indicates limited competition for VK2809, though there are several other drugs in development from companies like Eli Lilly, suggesting that competition may increase over time.
- **Treatment-eligible NASH population in the US.** The treatable population of Rezdiffra is used as a proxy for VK2809 since both drugs are selective TRβ agonists. Estimates for the treatment-eligible population of Rezdiffra among adults in the US vary from 1,255 to 1,699 adult patients out of a 1-million commercial-member population [17]. According to the US Census Bureau, roughly 92% of Americans had health insurance in 2023 [18] and since there are roughly 260 million adults in the US, this translates to an approximate commercial-member population of 239 million adults in the US. This means the treatment-eligible population of VK2809 can be approximated as 300,000 to 405,000 patients. To be conservative, 325,000 patients is taken as the estimate for the treatment-eligible population of VK2809.
- **Patient market share.** With only one approved therapy currently available, VK2809 may capture a generous portion of the NASH treatment market. However, given the pipeline of therapies under development, it is reasonable and fair to assume that there will be more alternatives than just Rezdiffra and VK2809 in the future. Thus, a conservative patient market share of 10% is assumed for VK2809.
- **Annual revenue per patient.** To estimate the annual revenue obtained for treating one patient with VK2809, the annual wholesale acquisition cost (WAC) of Rezdiffra is used as a proxy. The annual WAC of Rezdiffra is $47,400 USD [19] and assuming that VK2809 would sell at a 20% discount yields an annual revenue per patient of $38,000 USD.

Given that the treatment-eligible population for patients with NASH is roughly 325,000, the 10% market share estimate implies that VK2809 will treat roughly 32,500 patients annually. Multiplying this number with the estimated annual revenue per patient for VK2809 of $38,000 USD yields peak sales of $1.235 billion USD.

Valuation Model

Given the assumptions stated in the commercial model and market analysis, the probability-adjusted peak profit for VK2809 is projected to be $296 million USD (Table 1), expected to occur 11 years from 2025. Applying the pNPV framework outlined above results in an implied valuation of $10.24 USD per share for VK2809.

Table 6. Sensitivity analysis for pNPV per share of VK2809

		Years to peak sales				
		9	10	11	12	13
	741	8.12	7.06	6.14	5.34	4.65
	988	10.83	9.42	8.19	7.12	6.19
Peak sales (USDm)	**1235**	13.54	11.77	**10.24**	8.90	7.74
	1482	16.25	14.13	12.29	10.68	9.29
	1729	18.96	16.48	14.33	12.46	10.84

To obtain a valuation range, a sensitivity analysis was performed by varying both the peak sales and years to peak sales, as shown in Table 6. Given the peak sales base-case estimate of $1.235 billion USD is conservative, the range considered for valuation spans from this figure to 20% higher or $1.482 billion USD. Since the average development timelines were used to estimate time to peak sales, the sensitivity range adjusts the value by $\pm$ 1 year, resulting in a window of 10 to 12 years. Under these scenarios, the current valuation of VK2809 ranges from $8.90 to $14.13 USD per share.

Bibliography

[1] K. D. Keegan, *Biotechnology Valuation: An Introductory Guide*, 1st ed. Hoboken, NJ: John Wiley & Sons, 2008.

[2] A. Damodaran. "PE Ratio by Sector (US)." https://pages.stern.nyu.edu/~adamodar/New_Home_Page/datafile/pedata.html (accessed June 28, 2025).

[3] "VK2735 (Subcutaneous & Oral Formulations) — Metabolic Disease Program." Viking Therapeutics. https://vikingtherapeutics.com/pipeline/metabolic-disease-program/vk2735/ (accessed July 1, 2025).

[4] S. D. Emmerich, C. D. Fryar, B. Stierman, and C. L. Ogden, "Obesity and Severe Obesity Prevalence in Adults: United States, August 2021-August 2023," (in eng), *NCHS Data Brief,* no. 508, Sep 2024, doi: 10.15620/cdc/159281.

[5] I. Zaffina, M. C. Pelle, G. Armentaro, F. Giofrè, V. Cassano, A. Sciacqua, and F. Arturi, "Effect of dual glucose-dependent insulinotropic peptide/glucagon-like peptide-1 receptor agonist on weight loss in subjects with obesity," (in eng), *Front Endocrinol (Lausanne),* vol. 14, p. 1095753, 2023, doi: 10.3389/fendo.2023.1095753.

[6] T. D. Müller *et al.*, "Glucagon-like peptide 1 (GLP-1)," (in eng), *Mol Metab,* vol. 30, pp. 72-130, Dec 2019, doi: 10.1016/j.molmet.2019.09.010.

[7] "Viking Therapeutics Announces Positive Top-Line Results from Phase 2 VENTURE Trial of Dual GLP-1/GIP Receptor Agonist VK2735 in Patients with Obesity." Viking Therapeutics. https://ir.vikingtherapeutics.com/2024-02-27-Viking-Therapeutics-Announces-Positive-Top-Line-Results-from-Phase-2-VENTURE-Trial-of-Dual-GLP-1-GIP-Receptor-Agonist-VK2735-in-Patients-with-Obesity (accessed July 1, 2025).

[8] "Viking Therapeutics Reports New Data from VK2735 Obesity Program at ObesityWeek® 2024." Viking Therapeutics. https://ir.vikingtherapeutics.com/2024-11-04-Viking-Therapeutics-Reports-New-Data-from-VK2735-Obesity-Program-at-ObesityWeek-R-2024 (accessed July 1, 2025).

[9] "Viking Therapeutics Announces Results from Phase 1 Clinical Trial of Oral Tablet Formulation of Dual GLP-1/GIP Receptor Agonist VK2735." Viking Therapeutics. https://ir.vikingtherapeutics.com/2024-03-26-Viking-Therapeutics-Announces-Results-from-Phase-1-Clinical-Trial-of-Oral-Tablet-Formulation-of-Dual-GLP-1-GIP-Receptor-Agonist-VK2735 (accessed July 1, 2025).

[10] "GLP-1 Market: The Pipeline Expands (Global Sales of GLP-1)." TD Securities. https://www.tdsecurities.com/ca/en/glp1-market-the-pipeline-expands (accessed July 1, 2025).

[11] "U.S. GLP-1 Receptor Agonist Market Outlook." Grand View Research. https://www.grandviewresearch.com/horizon/outlook/glp-1-receptor-agonist-market/united-states (accessed July 1, 2025).

[12] L. C. Doward *et al.*, "Development of a Patient-Reported Outcome Measure for Non-Alcoholic Steatohepatitis (NASH-CHECK): Results of a Qualitative Study," *The Patient - Patient-Centered Outcomes Research,* vol. 14, no. 5, pp. 533-543, 2021/09/01 2021, doi: 10.1007/s40271-020-00485-w.

[13] Z. M. Younossi, A. B. Koenig, D. Abdelatif, Y. Fazel, L. Henry, and M. Wymer, "Global epidemiology of nonalcoholic fatty liver disease-Meta-analytic assessment of prevalence, incidence, and outcomes," (in eng), *Hepatology,* vol. 64, no. 1, pp. 73-84, Jul 2016, doi: 10.1002/hep.28431.

[14] E. Harris, "FDA Okays First Drug for Scarring From Fatty Liver Disease," (in eng), *Jama,* vol. 331, no. 17, p. 1439, May 7 2024, doi: 10.1001/jama.2024.5154.

[15] F. Saponaro, S. Sestito, M. Runfola, S. Rapposelli, and G. Chiellini, "Selective Thyroid Hormone Receptor-Beta (TRβ) Agonists: New Perspectives for the Treatment of Metabolic and Neurodegenerative Disorders," (in English), *Frontiers in Medicine,* Review vol. Volume 7 - 2020, 2020-July-09 2020, doi: 10.3389/fmed.2020.00331.

[16] "Viking Therapeutics Presents Results from Phase 2b VOYAGE Study of VK2809 in Biopsy-Confirmed NASH (MASH) at the 75th Liver Meeting® 2024." Viking Therapeutics Investor Relations.

https://ir.vikingtherapeutics.com/2024-11-19-Viking-Therapeutics-Presents-Results-from-Phase-2b-VOYAGE-Study-of-VK2809-in-Biopsy-Confirmed-NASH-MASH-at-the-75th-Liver-Meeting-R-2024 (accessed June 28, 2025).

[17] J. Fishman, Y. Kim, M. R. Charlton, Z. J. Smith, T. O'Connell, and E. M. Bercaw, "Estimation of the Eligible Population For Resmetirom Among Adults in the United States for Treatment of Non-Cirrhotic NASH with Moderate-to-Advanced Liver Fibrosis," (in eng), *Adv Ther,* vol. 41, no. 11, pp. 4172-4190, Nov 2024, doi: 10.1007/s12325-024-02989-5.

[18] U. S. C. Bureau, "Health Insurance Coverage in the United States: 2023," Washington, D.C., 2024. [Online]. Available: https://www.census.gov/library/publications/2024/demo/p60-284.html

[19] L. Leo and M. E. Sunny. "US FDA approves first drug for fatty liver disease NASH." Reuters. https://www.reuters.com/business/healthcare-pharmaceuticals/us-fda-approves-first-drug-fatty-liver-disease-nash-2024-03-14/ (accessed June 28, 2025).